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Phase 2 Presentation of Major Project

Implementation Stage

AY : 2025-26

“Knack: A Gamified Platform for Skill Development”

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Items 1 to 5 carry forward from Phase 1. Items 6 to 25 are new to this stage.

INTRODUCTION

- **Overview:** Knack is an AI-powered web platform that develops essential soft skills through interactive, scenario-based learning, delivered as a five-minute daily habit.
- **Current Scenario:** Engineering graduates often possess strong technical knowledge but lack "job-readiness" due to a deficit in soft skill training.
- **Relevance:** Soft skills like communication and situational judgment are the most sought-after traits by global employers.
- **Status at Phase 2:** The platform is **fully implemented and running** as a **career simulation**. The trainee is not asked questions about a workplace; they are placed inside one. They sit a group discussion, clear five interview rounds, join a company, work there through timed days made of messages, decisions and live conversations with colleagues who remember them, and finish with a performance review, a salary offer and a negotiation.
- **What is measured:** ten professional attributes on a 0-100 scale that **fall as well as rise**, kept deliberately separate from XP. XP is what a trainee earns for turning up; an attribute is a claim about what they can actually do.

PROBLEM STATEMENT

The Core Problem: Over 60% of fresh graduates in India lack the soft skills required for the professional workforce.

Key Issues:

- **Educational Gap:** Soft skills remain largely untaught in standard engineering curricula.
- **Passive Learning Failure:** Existing solutions like workshops or MOOCs are passive and non-engaging, leading to low completion rates.
- **Measurement Difficulty:** Traditional methods provide no credible way to measure or track situational judgment growth.
- **Cost of Personalisation:** An AI tutor that remembers a learner normally grows more expensive with every session, which makes daily use unaffordable at scale.

OBJECTIVES

- **Bridge Readiness Gaps:** Align academic learning with professional situational demands.
- **Dynamic Learning Environments:** Deliver adaptive, real-world professional challenges generated for the individual learner.
- **Sustained Habit Formation:** Turn soft-skill development into a daily ritual using XP, levels and streaks.
- **Measurable Progress Tracking:** Visualise qualitative growth through per-track skill scores and a composite Professional Readiness Score.
- **Bounded Operating Cost:** Keep the AI prompt a constant size no matter how long a learner has been training. (Added at implementation stage.)
- **Graceful Degradation:** Remain usable when the free AI service is rate-limited. (Added at implementation stage.)
- **Assess Recognition and Production Separately:** Test both the ability to choose the right move and the ability to write it. (Added at implementation stage.)
- **Rehearse the Real Gate:** Give the learner a graded mock interview drawn from their own resume, since the placement interview is what the training is ultimately for. (Added at implementation stage.)
- **Teach Through Consequence, Not Correction:** Place the learner inside a workplace where decisions affect relationships, opportunities and pay, and let those consequences arrive later than the decision that caused them. (Added at implementation stage.)

LITERATURE SURVEY

Paper Title	Key Findings	Limitations
Effectiveness of Gamified Language Learning (Duolingo Research, 2020)	Bite-sized lessons and daily streaks are highly effective for long-term retention.	Focused on language acquisition rather than complex professional situational judgment.
Employability and Skill Gap Study (NASSCOM, 2023)	Over 60% of engineering graduates lack soft skills despite strong technical skills.	Statistical overview of the problem; proposes no technological solution.
Impact of Gamification on Learning Outcomes (Lampropoulos et al., 2024)	Gamified learning consistently outperforms traditional methods in engagement and performance.	Highly context-dependent; requires thoughtful pedagogical design to sustain interest.
Research AI: Integrating AI and Gamification (Marengo et al., 2025)	AI personalises educational pathways; gamification supports interactive soft-skill assessment.	Raises data privacy and accessibility challenges in resource-limited institutions.
Judging LLM-as-a-Judge (Zheng et al., 2023)	Language models can apply a written rubric to open-ended answers with useful agreement.	Models show leniency bias and sensitivity to answer length; needs calibrated prompting.
Generative Agents (Park et al., 2023)	Periodically rewriting a compact memory keeps long-running AI agents coherent over time.	Demonstrated in simulation, not in an assessed educational setting.

The last two rows were added at Phase 2; they directly informed the grading and memory design.

RESEARCH GAP

- **Passive vs. Active Learning:** Existing platforms rely on passive video consumption, which lacks the active engagement necessary for behavioural change.
- **Lack of Real-Time Feedback:** Curriculum-based training gives no immediate, AI-driven coaching on open-ended situational responses.
- **Stagnant Content:** Most workshops use fixed scenarios that do not adapt to an individual student's performance or skill level.
- **Measurement Infrastructure:** There is a significant gap in measuring and visualising soft-skill growth through quantitative metrics.
- **Unbounded Memory Cost:** Systems that personalise from history typically inject past interactions into every prompt, so cost and latency grow without limit. No reviewed work addresses this for a daily-use educational tool.

EXISTING SYSTEM vs PROPOSED SYSTEM

Aspect	Existing Systems	Proposed System (Knack)
Response format	Multiple choice or select-an-option	Free-text; the learner writes the reply they would actually send
Content source	Fixed bank, identical for every learner	Generated per learner by a large language model
Personalisation	None, or level selection only	Stated profile + GitHub portfolio + demonstrated weaknesses
Feedback	Correct / incorrect, or a score alone	Score 1–10 with strengths, gaps and one concrete improvement
Adaptation	Static difficulty	Difficulty tracks a rolling score and its recent trend
Memory of the learner	None	A rewritten summary carried forward at constant cost
Availability	Fails when the service is down	Falls back to a local scenario bank, clearly labelled

SYSTEM REQUIREMENTS

Hardware Requirements

Processor	Dual-core x86-64, 2.0 GHz or above
Memory	4 GB minimum, 8 GB recommended
Storage	500 MB including dependencies
Network	Internet for AI calls; app runs without it
Client device	Any modern browser, phone or laptop

Functional Requirements

- Account creation and secure login
- Profile capture and editing
- Scenario generation and response grading
- XP, level, streak and skill tracking
- History with full attempt replay

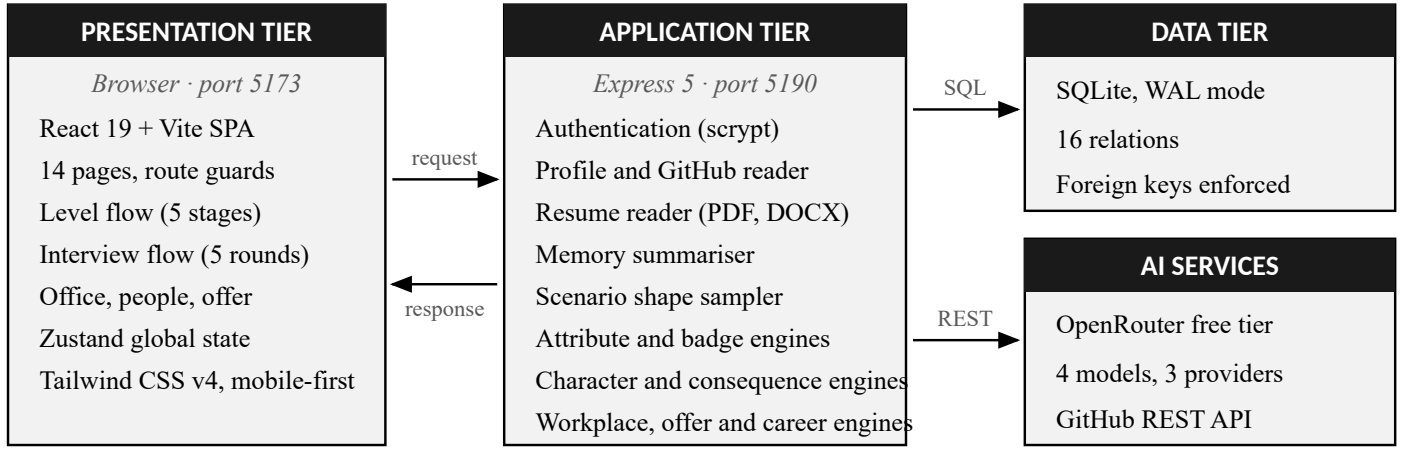
Software Requirements

OS	Windows 10/11, Linux or macOS
Runtime	Node.js v24 LTS
Database	SQLite (file-based, no server)
AI service	OpenRouter account and API key
Browser	Chrome, Edge or Firefox, current version

Non-Functional Requirements

- **Bounded cost** — prompt size independent of history
- **Security** — memory-hard password hashing
- **Isolation** — data scoped to the owning account
- **Usability** — one primary action per screen
- **Accessibility** — respects reduced-motion setting

METHODOLOGY / PROPOSED ARCHITECTURE



The application tier binds the loopback interface only, so the resume, the database file and the key are never reachable from the network, even while the app is open on a phone over Wi-Fi.

Figure 1. Three-tier system architecture

SYSTEM DESIGN AND FLOW DIAGRAM

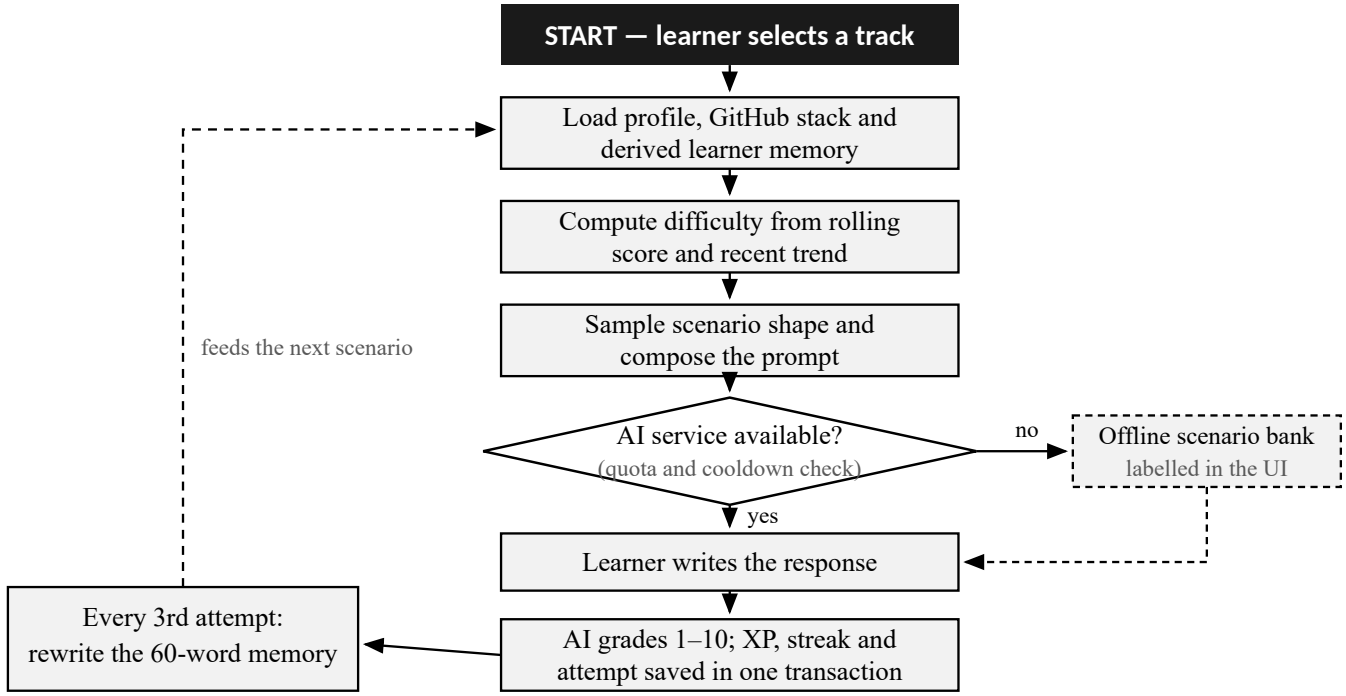
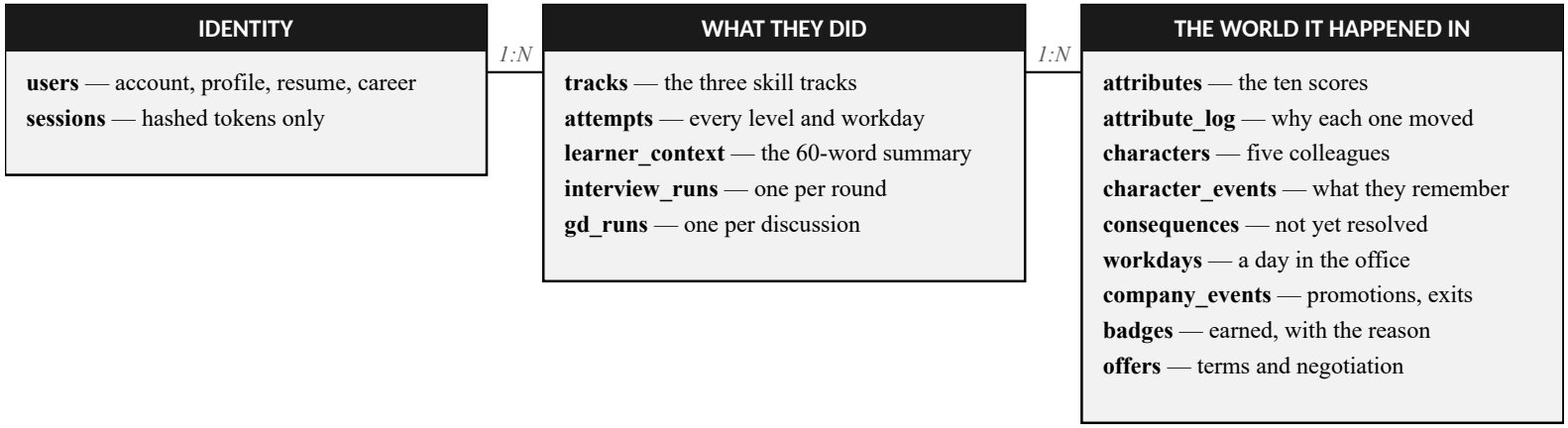


Figure 2. End-to-end execution flow. The memory rewrite runs after the response is returned, so it never adds to the learner's waiting time.

DATABASE DESIGN



16 relations, SQLite in WAL mode, foreign keys enforced. Every child row cascades from the account, so deleting a trainee removes their whole simulation.

Two design rules: history is compressed rather than accumulated, and an interview or discussion is kept apart from attempts so it can never move a track score.

Figure 3. Schema by purpose — identity, the training record, and the simulated world

MODULES AND THEIR FUNCTIONALITY

Module	Functionality	Key design decision
Authentication	Registration, login, sessions, protected routes	Only the hash of a session token is stored, so a database leak grants no access
Profile	Eight onboarding questions and their validation	Defined once on the server so the form and validation cannot diverge
GitHub Reader	Reads public repositories, derives the learner's real stack	Forks and archived repositories excluded — not evidence of the learner's own work
Resume Reader	Extracts text from an uploaded PDF, DOCX or TXT and distils it	Written against zlib, already in Node — no parser dependency, and a scanned PDF is reported as unreadable rather than stored as noise
Level Engine	Builds four judgement calls plus one written stage, then grades them	Option scores never reach the browser; the level is held server-side and only the choices are submitted
Interview Engine	Five rounds of mock interview questions and a graded debrief	The next round unlocks on performance, not on XP, so a round cannot be ground through
Attribute Engine	Ten permanent 0-100 attributes, their tiers and their movement	Every delta is capped and written with a reason, so no single answer is decisive and every change can be explained rather than merely displayed

The seven modules above carry the platform. The seven overleaf carry the career simulation.

MODULES — THE SIMULATION LAYER

Module	Functionality	Key design decision
Character Engine	Five persistent colleagues, their relationships and their memory	Stores what a colleague remembers, not just a score — a number cannot produce the sentence "you helped me when I first joined"
Consequence Engine	Schedules and later fires the results of notable decisions	Due counts are in completed levels rather than days: the world moves when the trainee does, not while the app is closed
Discussion Engine	Group discussion with five AI candidates, beat by beat	Volume is explicitly not rewarded; a candidate who never speaks is capped at 25 on every dimension and told why
Workplace Engine	Rooms, timed beats, live dialogue and the daily summary	Each beat is generated after the previous one resolves, so the day can follow from itself rather than being written all at once
Career and Offer Engine	Phases, compensation, negotiation and promotion	Every gate is a competency check; promotion needs Leadership, not attendance
Memory Controller	Rewrites the learner summary every three attempts	Runs after the response is sent, so it never adds to perceived latency
Scenario Sampler	Selects the shape of each scenario before generation	18,480 combinations generated by the system, not requested from the model
Prompt Templates	Eight templates: scenario, grade, level, level grade, resume, interview, debrief, memory	Isolated in one file so they can be tuned without touching route logic
AI Adapter	Model calls, retries, cross-provider fallback chain	A rate-limit response is never retried; reset timestamps are honoured
Gamification Engine	XP, levels, streaks, skill scores, readiness score	Self-rated confidence sets starting difficulty only, never the score
Offline Fallback	Local scenario generator and deterministic rubric	Always labelled in the interface, never presented as AI output

TECHNOLOGIES, TOOLS AND SOFTWARE USED

Layer	Technology	Purpose
Frontend	React 19, Vite 7, Tailwind CSS v4, Framer Motion, Zustand, React Router 7	Single-page interface, state, animation and routing
Backend	Node.js 24, Express 5, cookie-parser	REST endpoints, middleware and session handling
Security	scrypt from Node's built-in crypto module	Memory-hard password hashing without an extra dependency
Database	SQLite via better-sqlite3, WAL mode	Local transactional storage, sixteen relations
Document parsing	zlib from Node's standard library	PDF and DOCX text extraction with no third-party parser
AI engine	A four-model fallback chain on OpenRouter's free tier (MiniMax M2.7, Nemotron 3 Super, Gemma 4 31B, Gemma 4 26B)	Scenario and question generation, grading, resume, portfolio and memory summarisation
External API	GitHub REST API v3	Reads public repository metadata for portfolio grounding
Tools	Git and GitHub, VS Code, Chrome DevTools	Version control, development and mobile testing

Complete source code: **github.com/ishaaqdev/Knack**

DEVIATIONS FROM PHASE 1

Area	Phase 1 proposal	Implemented	Reason for the change
Frontend	Flutter (mobile and web)	React 19 + Vite, responsive web	One responsive codebase reaches phone and laptop without a second build toolchain
Backend	Node.js with FastAPI (Python)	Node.js with Express 5 only	No Python component proved necessary; two runtimes would double deployment complexity
AI model	Gemini API or GPT-4o	Gemma 4 31B via OpenRouter	Both proposed models are paid. OpenRouter's free tier made daily use possible at zero cost
Gamification store	Redis for live leaderboards	SQLite; leaderboards removed	Ranking classmates on communication ability would deter the students who most need practice
Streak triggers	Firebase	Calendar-date logic in SQLite	A single local database removes an external dependency for a purely date-based rule
Tracks	Four, including Situational Judgment	Three	Situational judgment is exercised inside all three tracks rather than isolated as its own

The six above reduced external dependencies or running cost.

DEVIATIONS — SCOPE AND ADDITIONS

Area	Phase 1 proposal	Implemented	Reason for the change
Assessment format	One written scenario per session	A level: four multiple-choice calls plus one written stage	Choosing the right move and being able to write it are different skills. One question measured only the second and took only one reading of one situation
Learner input	Onboarding form and GitHub	Resume upload added as a third source	The resume already holds the projects and numbers the form asks for, and it is the document a real interviewer would have read
Interview practice	Not in scope	Five graded mock-interview rounds	The placement interview is the gate the training exists for; practising the skills without ever rehearsing the gate leaves the loop open
Scope	A gamified quiz over three tracks	A six-phase career simulation	The design document's central argument: questions about a workplace teach less than being placed inside one. The tracks survive as the practice mode
What is measured	XP, levels, per-track scores	Ten attributes that also fall, kept separate from XP	XP measures attendance. Separating the two is what stops the app rewarding a trainee who turns up daily and learns nothing
Badges, skill tree	Planned	24 badges built; skill tree deferred	Badges became meaningful once there were phases and relationships to attach them to

Rows 1 to 6 reduced external dependencies or running cost. The rest are additions made once the core loop worked and the scope was widened to the full design document. No proposed objective was dropped.

IMPLEMENTATION DETAILS — PERSONALISATION

Four independent signals are combined into every scenario. Each is bounded in size by construction, which is what keeps the prompt a constant length.

- **Stated profile (8 fields):** academic year, branch, target role, prior team experience, preferred work setting, self-identified difficulties and confidence per track. Collected once at onboarding, editable later.
- **GitHub portfolio (optional):** reads the learner's public repositories, discards forks and archived work, and reduces them to at most six technologies and three problem domains.
- **Resume (optional):** the uploaded document is reduced to at most ten technologies, four projects and three factual claims. Only the technology list reaches a scenario prompt; the projects and claims are held for the interviewer, who is the one who should be asking about them.
- **Derived memory:** a 60-word summary of what past attempts revealed, rewritten by the model every three attempts from a six-attempt window.
- **Scenario shape sampling:** the counterpart, the medium, the moment and the pressure are chosen by the server from **18,480 combinations** and passed to the model as requirements, rather than asking the model to vary its own output.

The prompt explicitly forbids restating the learner's profile in the scenario text — the background selects the setting and the stakes, and must never appear in the situation itself.

IMPLEMENTATION DETAILS — BOUNDED MEMORY

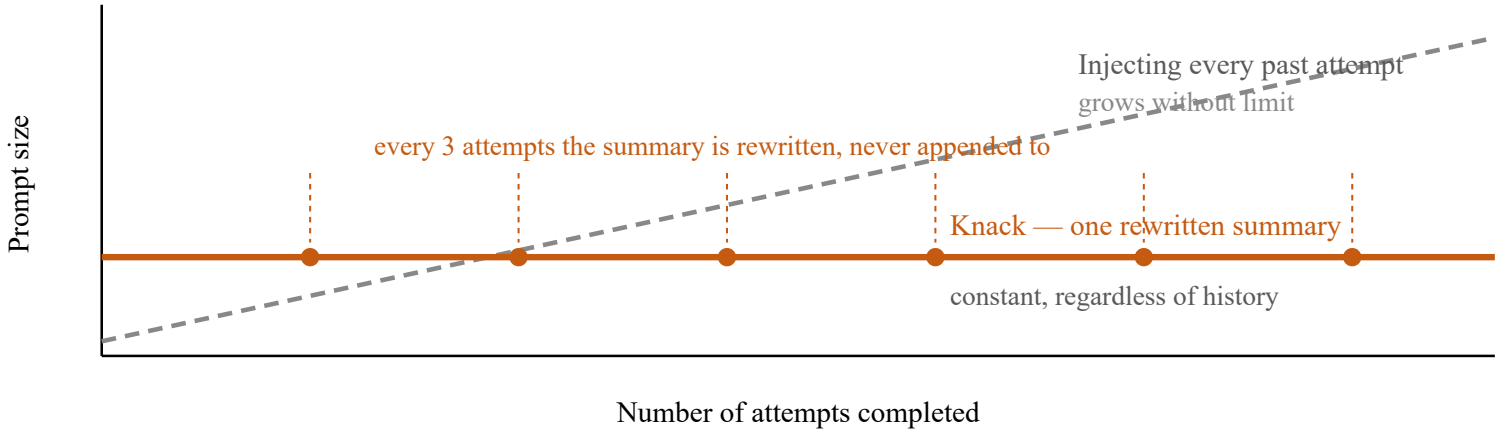


Figure 4. Prompt size against usage (design property, illustrated)

Observed effect. After three deliberately vague answers, the system recorded: *"consistently provides vague, non-committal responses… describes their intent to communicate rather than concrete actions."* On the fourth such answer the grader replied: **"This is the same gap as last time."**

LEVEL STRUCTURE AND INTERVIEW ROUNDS

A LEVEL — recognition, then production

STAGE 1	STAGE 2	STAGE 3	STAGE 4	STAGE 5 — WRITTEN
one moment, four responses 9 / 6 / 3 / 1	one moment, four responses 9 / 6 / 3 / 1	one moment, four responses 9 / 6 / 3 / 1	one moment, four responses 9 / 6 / 3 / 1	no options; write what you would actually send graded 1 to 10

40% of the level score · up to 25 XP each

60% · up to 100 XP

Every stage also offers "something else" — a free-text answer graded alongside the written stage, because a real workplace never presents four options.

THE INTERVIEW — five rounds, unlocked by performance

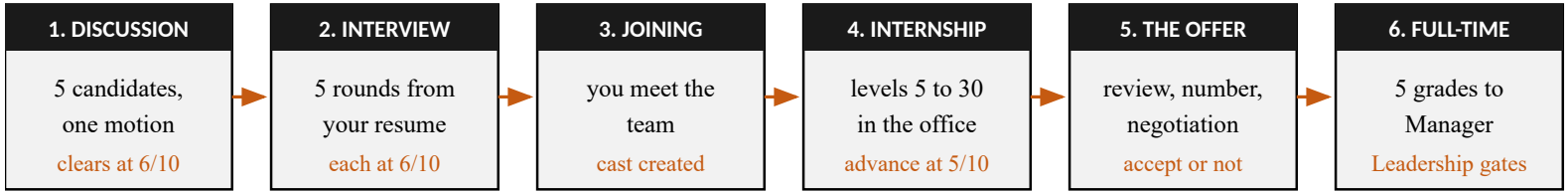
ROUND 1	ROUND 2	ROUND 3	ROUND 4	ROUND 5	
Introduction 3 questions	Behavioural 3 questions	Your work 3 questions	Pressure 3 questions	Closing 3 questions	Questions come from the resume, by name

A round is cleared at 6 out of 10. Clearing it opens the next one — XP alone never does, so a weak round repeated ten times leaves the candidate exactly where they started.

Figure 5. The two training loops added at implementation stage

THE CAREER SIMULATION

SIX PHASES — each one entered by clearing the last, never by earning XP

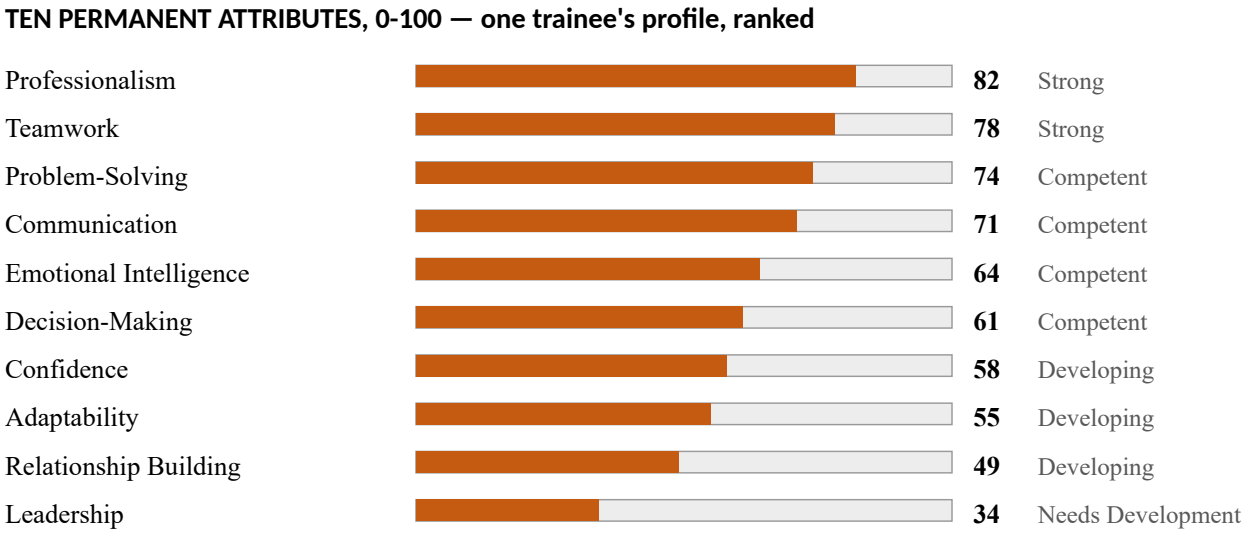


Why gate on performance rather than XP. XP is a currency you accumulate by turning up. A phase gate is a claim that the trainee can do the thing. Keeping them separate is what stops the simulation from rewarding attendance: a weak round repeated ten times leaves the candidate exactly where they started.

The world persists across all six. The colleagues met at phase three are the same people who appear in phase four, remember phase four in phase five, and decide whether anyone argues for the trainee's salary band when the offer is written.

Figure 6. The career spine. Nothing in it can be skipped or bought with XP.

PROFESSIONAL ATTRIBUTES



These are deliberately not XP, and they fall. XP measures attendance; an attribute is a claim about capability. A trainee can end a month with a great deal of XP and a worse Professionalism score than they started with, and the two numbers disagreeing is the point.

Measured: one discussion where the candidate supported the room but never led moved Teamwork +2.1, Emotional Intelligence +2.5, and Leadership -0.4.

Figure 7. Each attribute carries a tier and, below 60, the specific actions that move it

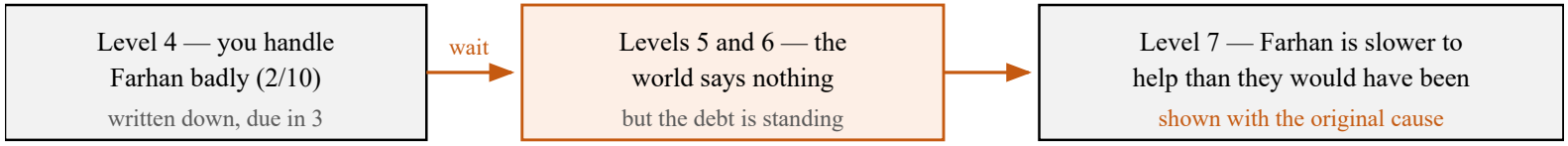
PEOPLE, MEMORY AND CONSEQUENCES

FIVE PERSISTENT COLLEAGUES — created the day the trainee joins

Hema	Arun	Kiran	Meera	Farhan
Manager	Senior Engineer	Fellow Intern	Product Manager	HR Partner
74 · Positive	41 · Neutral	88 · Trusted	52 · Neutral	30 · Distant

Each stores what they remember, not just a number.

THE CONSEQUENCE ENGINE — a decision finishes happening later than it is made



A cost paid at the moment of the decision is a score, not a consequence. The delay is what makes it one — far enough that it lands as a callback, near enough that the trainee still remembers what they did. The count is in completed levels, not days: the world moves when the trainee does. A grudge also reshapes the next situation instead of being announced.

Figure 8. Relationships persist; decisions come back

A DAY IN THE WORKPLACE

STEP 1 — CHOOSE A ROOM. Where you go decides what happens in the day.

Your desk	Manager's office	Meeting room	Cafeteria	HR

STEP 2 — THREE BEATS, EACH WRITTEN AFTER THE LAST ONE RESOLVES

A MESSAGE	→	A DECISION	→	A CONVERSATION
something needs a written reply 90 seconds		four options, scored 9 / 6 / 3 / 1 60 seconds		up to 3 live turns, they answer in character 120 seconds

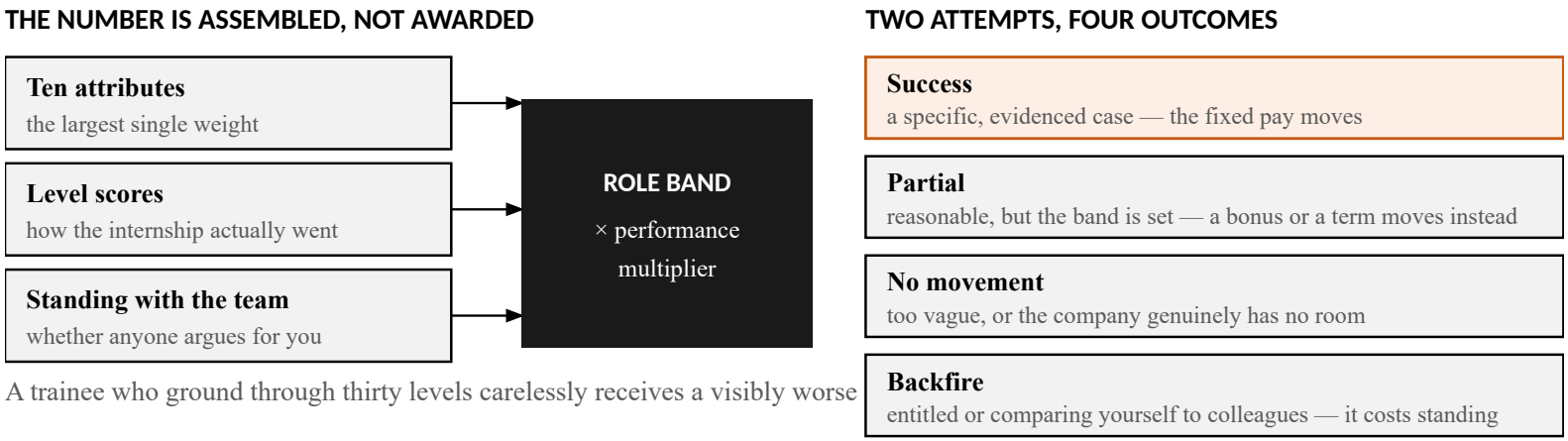
Running out of time is an answer, with its own consequence.

Three beats, not the nine-item timetable the design document sketches. A daily habit that takes forty minutes is a habit nobody keeps, and the same nine beats every day is exactly the linear course the project set out not to build.

Each beat is generated after the previous one is answered, with what already happened in the prompt — so a promise made at ten in the morning comes due at three in the afternoon.

Figure 9. The daily loop, and why it is short

THE OFFER AND NEGOTIATION

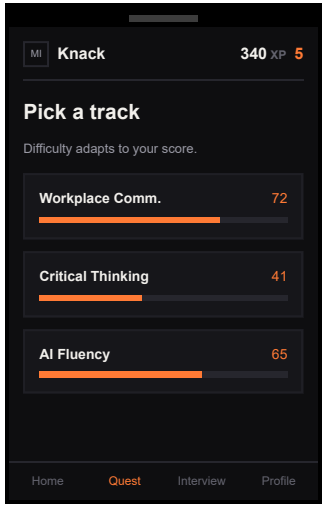


Only one of the four outcomes is "you got more money." A badly made ask costs standing with the manager who sat through it. Otherwise negotiating would be free — and the one thing every graduate needs to learn about it is that it is not.

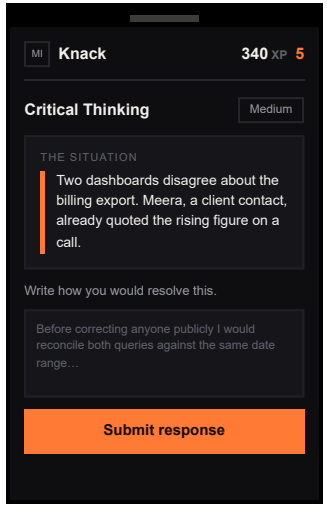
Measured: a vague ask scored 30/25/20 and moved nothing. The same trainee asking for a named figure with two pieces of evidence scored 75/80/70 — the grader clearly separates the two.

Figure 10. Compensation, and the conversation about it

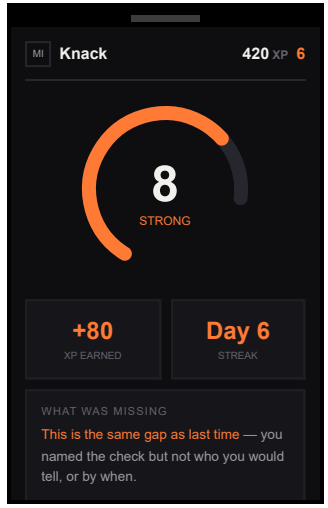
SCREENSHOTS — THE TRAINING PLATFORM



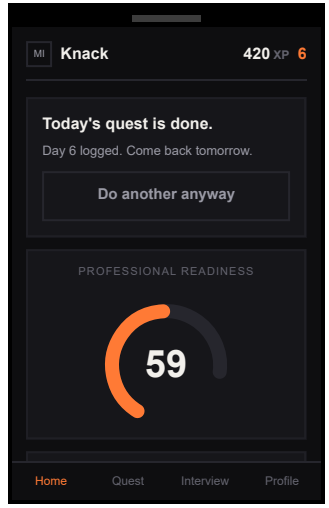
Track selection



Written stage



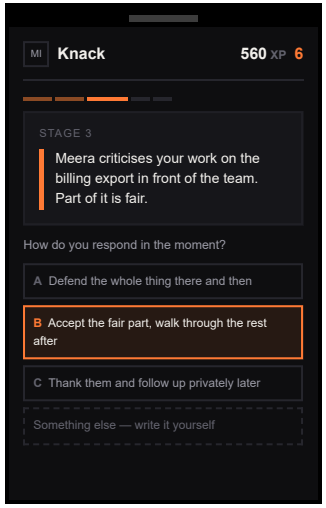
Level score



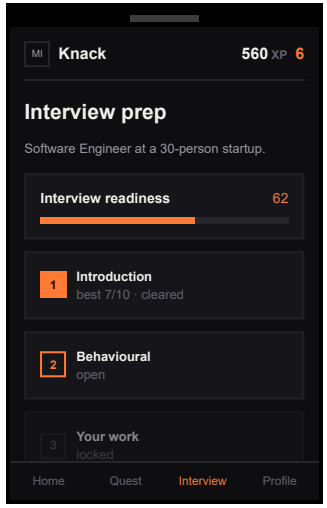
Dashboard

Mobile-first throughout. XP and streak stay visible on every screen; one primary action per page.

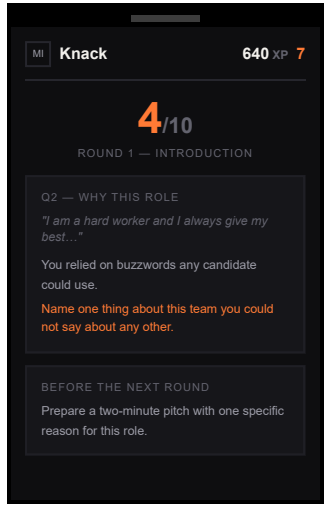
SCREENSHOTS — THE SIMULATION



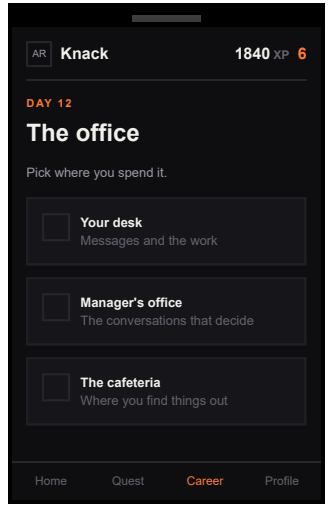
Judgement stage



Interview rounds



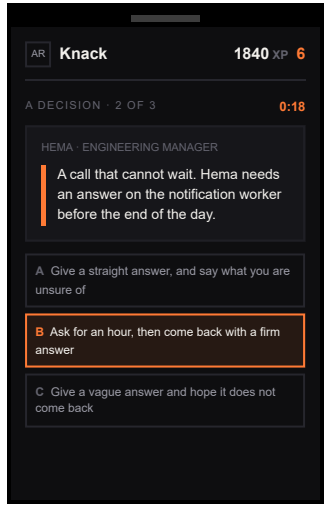
Interview debrief



The office



People and memory



A timed beat

The office, the people who remember you, and a decision on a timer.

RESULTS / CURRENT PROGRESS

Property tested	Method	Observed result
Grader discrimination	Four answers of differing quality on comparable scenarios	Dismissive one-line answer **1/10**; polite but non-committal **2/10**; reasonable **6/10**; specific and offering a real choice **9/10**
Profile sensitivity	Two profiles differing in every field, same track	CSE / startup received a payment-gateway code review; ECE / core / service company received a PCB layout error in a sprint review
Portfolio grounding	Public GitHub linked, 10 repositories read	Stack correctly derived; scenarios moved to model-latency and sensor-calibration disputes
Derived memory	Evasive answer pattern planted over three attempts	System identified the pattern unprompted and reported it as a recurring gap
Memory in feedback	Same evasive answer given a fourth time	Grader replied **"This is the same gap as last time"**
Data isolation	Second account created while the first held attempts	Account B saw 0 quests; account A saw 4

Personalisation and assessment, verified live. The simulation results follow.

RESULTS — THE SIMULATION

Property tested	Method	Observed result
Scenario variety	Six generations, same track and difficulty	Before the fix: 7/9 distinct, 2 identical pairs. After: **6/6 distinct**, no duplicates, six different counterpart roles
Resume extraction	Text pulled from an uploaded document and distilled	10 skills, 3 projects and 3 factual claims recovered correctly; the projects were named back as written, and a document with no text layer is refused rather than stored
Level scoring	A level answered with mixed-quality choices	Chosen options scored **9, 1, 8, 9** against a written stage of **7**; level score 7, 139 XP. The free-text answer given instead of an option was graded **8/10**
Interview gating	Round 1 answered strongly, weakly, and not at all	Round scored **4/10**, below the 6/10 gate, and round 2 stayed locked. Verdict named the unanswered question as the reason
Attributes disagree with XP	A group discussion where the candidate supported others but never led	Teamwork **+2.1**, Emotional Intelligence **+2.5**, Leadership **-0.4** from one performance. XP rose throughout. This is the behaviour a single score cannot produce
Discussion assessment	Two well-timed contributions, two deliberate silences	Scored 7/10 and cleared. Listening 85, Teamwork 80, Leadership 45 — the assessor identified that the candidate built on others but never shaped direction
Day chaining	Three beats in one workplace day	Beat 2 opened *"You are already behind on the code review you promised to finish by noon"* — a commitment made in beat 1. The model never sees the whole day at once
Negotiation discrimination	The same trainee asking vaguely, then specifically	Vague ask scored 30/25/20 and moved nothing. Named figure with two pieces of evidence scored **75/80/70**; it still moved nothing, because the company had no room at that performance — which is the honest outcome

Stated honestly: assessment uses a single grader that has not been checked against human experts, and testing used profiles we constructed rather than a student cohort. What is demonstrated is that the grader *discriminates*, not that it is correct.

CHALLENGES FACED AND SOLUTIONS ADOPTED

Challenge	Cause identified	Solution adopted
Free AI quota exhausted within hours	OpenRouter allows 50 free requests per day. Our retry policy used up to six requests per scenario.	One attempt per model, rate limits never retried, and reset timestamps honoured through a cooldown
The fallback model failed identically	Both models in the chain were served by the same upstream provider.	Chain rebuilt across three independent providers
Every scenario read the same	Instructing the model to vary its output changed the wording but not the structure.	18,480 scenario shapes sampled by the server and passed as requirements
Model fixated on one project	The GitHub summary named a specific project and the model anchored on it.	Summary kept for display; only the technology list is sent to the model
Profile read back in the scenario	Scenarios opened "As a final-year CSE student aiming for…".	Explicit rule: background sets the scene and must never appear in it

Five faced while building the platform. Five more overleaf, from the simulation work.

CHALLENGES — THE SIMULATION STAGE

Challenge	Cause identified	Solution adopted
Some models returned no output	Nearly every free model now reasons before answering, and those hidden tokens are charged against the output ceiling. One spent 2,691 of them and returned nothing.	Reasoning trace suppressed, effort lowered, and a fixed allowance added to every ceiling — a ceiling costs nothing when the model stops early
Feedback landed on the wrong answer	The grader numbered its per-answer critiques from one while the code read them from zero, so every candidate saw criticism of a paragraph they had not written.	Index base detected from the returned values; unanswered questions graded locally, never sent to the model
Attribute changes were silently discarded	The ten attributes were seeded on the dashboard read, so a trainee who went from onboarding straight into a level had no attribute rows at all, and every delta written against them updated nothing.	Seeding moved to the gameplay gate, where nothing can score before the thing it scores into exists. Found by an end-to-end run in which a 9/10 moved a relationship but no attributes
The review invented a colleague	The end-of-internship review produced "Jordan has spoken up for you" on a team with no Jordan on it, despite the prompt forbidding it.	Any sentence naming someone not on the team is dropped before display; a rule in a prompt is a request, and this one mattered too much to be one
A level took longer than anyone would wait	Two model calls issued together queue behind each other on a free provider.	The written stage is started when the level is served and collected while the trainee answers the four choices, so it is ready before they reach it
Port silently taken by another service	The app worked only because the hostname resolved over IPv6.	Moved to a free port and bound the loopback interface explicitly

FUTURE ENHANCEMENTS

- **Validate the grader against human raters.** Collect expert scores for the same responses and measure agreement. This converts the present result — that the grader separates good answers from poor ones — into a claim that it is correct. Highest value next step.
- **Two-grader reconciliation.** Sample two independent gradings and reconcile where they diverge, reducing variance at roughly double the inference cost.
- **Spoken interview rounds.** The five written rounds already exist; adding audio capture and transcription would put them through the same grader in the medium a real interview actually uses.
- **Group discussion round.** Several AI candidates with different personalities, scored on when the learner enters the discussion rather than how much they say — the one placement stage the current build does not rehearse.
- **Instructor dashboard.** Anonymised, class-level weakness aggregation so a department can see which competencies to address in teaching.
- **Achievement badges**, deferred from Phase 1 and compatible with the existing progression model.
- **Account recovery and a native mobile client.** The interface is already responsive, so a mobile client is an incremental rather than architectural change.

CONCLUSION

- **Objectives met.** A working platform delivers AI-generated, personalised workplace scenarios, grades free-text responses on a calibrated rubric, and tracks progress through XP, levels, streaks and a Professional Readiness Score.
- **Personalisation is demonstrable.** Substituting the learner profile changes the scenario substantially; a linked GitHub account grounds scenarios in the technologies the learner actually uses; and an uploaded resume lets the mock interviewer ask about their projects by name.
- **Two skills are measured, not one.** A level asks the learner to recognise the right move four times and then to write it once, and weights the writing higher — recognising good communication and producing it are not the same ability.
- **The simulation is the contribution.** Ten attributes that fall as well as rise, colleagues who remember specific interactions, consequences that arrive two or three levels after the decision that caused them, and a salary negotiation with four possible outcomes. Together these produce something a quiz cannot: a trainee who accumulates XP while their Leadership score falls, and can see exactly why.
- **The principal contribution is bounded memory.** By rewriting a short summary every three attempts rather than accumulating history, the system carries forward what it has learned at a constant cost. Its effect is visible in feedback that says **"this is the same gap as last time"** — a statement a stateless system cannot produce.
- **Limitations are acknowledged.** The grader is unvalidated against human experts, testing used synthetic profiles, and no longitudinal study of learning outcomes has been carried out.

Complete implementation and documentation: **github.com/ishaaqdev/Knack**

REFERENCES

1. **Deterding, S., et al. (2011).** From Game Design Elements to Gamefulness: Defining Gamification. *Proc. 15th International Academic MindTrek Conference.*
2. **Kapp, K. M. (2012).** *The Gamification of Learning and Instruction.* Wiley Publishing.
3. **Duolingo Research (2020).** Effectiveness of Gamified Language Learning and Long-term Retention Strategies. *Journal of Computer Assisted Learning.*
4. **Altomari, L., et al. (2023).** Gamification in Soft Skills Assessment: Trends and Opportunities. *IEEE / Educational Research Papers.*
5. **Thomas, M. (2023).** AI Chatbots in Mobile Learning Environments: A Qualitative Study. *International Journal of Educational Technology.*
6. **World Economic Forum (2023).** The Future of Jobs Report 2023. *WEF Strategy Papers.*
7. **Lampropoulos, G., et al. (2024).** Impact of Gamification on Learning Outcomes: A Systematic Review. *Journal of Interactive Learning Environments.*
8. **Marengo, A., et al. (2025).** Integrating AI and Gamification for Professional Skill Development. *Proc. IEEE Int. Conf. on Learning Technologies.*

REFERENCES (CONTINUED)

9. **Mercer-Mettl (2025).** India Graduate Skill Index Report. *Industry Skill Assessment Report.*
10. **Reddy, V., & Sharma, P. (2025).** Bridging the Employability Gap Using AI-driven Soft Skill Training. *Journal of Vocational Education.*
11. **Zheng, L., et al. (2023).** Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. *Advances in Neural Information Processing Systems 36.*
12. **Park, J. S., et al. (2023).** Generative Agents: Interactive Simulacra of Human Behavior. *Proc. ACM UIST.*
13. **Lewis, P., et al. (2020).** Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *NeurIPS 33.*
14. **Hattie, J., & Timperley, H. (2007).** The Power of Feedback. *Review of Educational Research, 77(1).*
15. **Corbett, A. T., & Anderson, J. R. (1994).** Knowledge Tracing: Modeling the Acquisition of Procedural Knowledge. *User Modeling and User-Adapted Interaction, 4.*
16. **Percival, C., & Josefsson, S. (2016).** The scrypt Password-Based Key Derivation Function. *RFC 7914, IETF.*

Appendix A: Workplace Communication

Focus: Navigating professional hierarchy and clarity.

- **The Rebuttal:** You fundamentally disagree with your Team Lead's technical direction during a client briefing; deliver a professional rebuttal that protects project integrity without undermining authority.
- **The Credit Claim:** A peer takes credit for your code or logic in a team meeting; reclaim your contribution professionally so your performance is accurately recorded.
- **The Crisis Update:** A major deadline will be missed due to a server failure; draft a "no-excuse" update to your manager that proposes a recovery timeline.

Implemented: the system now generates scenarios of this kind automatically, varying the counterpart, the medium, the moment and the pressure on every generation.

Appendix B: Critical Thinking

Focus: Resource management and data-driven logic.

- **The Runway Decision:** Your startup has exactly ₹50,000 remaining; justify spending it on urgent lead generation or on fixing a critical security bug.
- **The Presentation Pivot:** During a project defence a panelist identifies a fundamental flaw in your logic; decide in real time whether to defend the architecture or pivot.
- **The Hiring Logic:** Evaluate two candidates with identical CGPAs on situational judgment data and justify your selection against a competency rubric.

Implemented: difficulty for this track is set from the learner's rolling average of the last five scores, shifting one band at a time as the trend moves.

Appendix C: Digital & AI Fluency

Focus: Critical oversight of automated systems.

- **The Hallucination Audit:** An AI-generated proposal contains two subtle factual errors about local regulations; identify them and evaluate the risk of sending the document.
- **The Component Design Challenge:** Generate boilerplate for an analytics dashboard with an AI tool, identify a logic flaw in the state management, and refine the prompt so the design is scalable and secure.
- **The Ethical Risk Assessment:** An organisation plans to automate all student counselling with language models; evaluate the top three ethical and privacy risks.

Implemented: this track grades whether the learner named a specific verification step, rather than whether they expressed general caution.